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"A Study on financial literacy and saving behaviour among students."

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Introduction

Financial literacy is an essential skill for students as they begin managing their own money. With the rise of digital payments and increasing expenses, students need basic knowledge of budgeting, saving, and responsible financial behaviour. However, many students still lack awareness about financial management, which affects their saving habits and spending patterns.

This study aims to examine students' financial literacy, their use of digital payments, and their saving behaviours through a survey. The findings will help understand how well students manage their finances and where improvements in financial education are needed.

Objectives of the Study

1. To assess the level of financial literacy among students.
2. To analyse students' saving habits and budgeting practices.
3. To examine spending patterns and the use of digital payment methods.
4. To evaluate students' awareness of basic investment options and financial concepts.
5. To identify the factors that influence students' financial behaviour.
6. To suggest measures for improving financial awareness and money-management skills among students.

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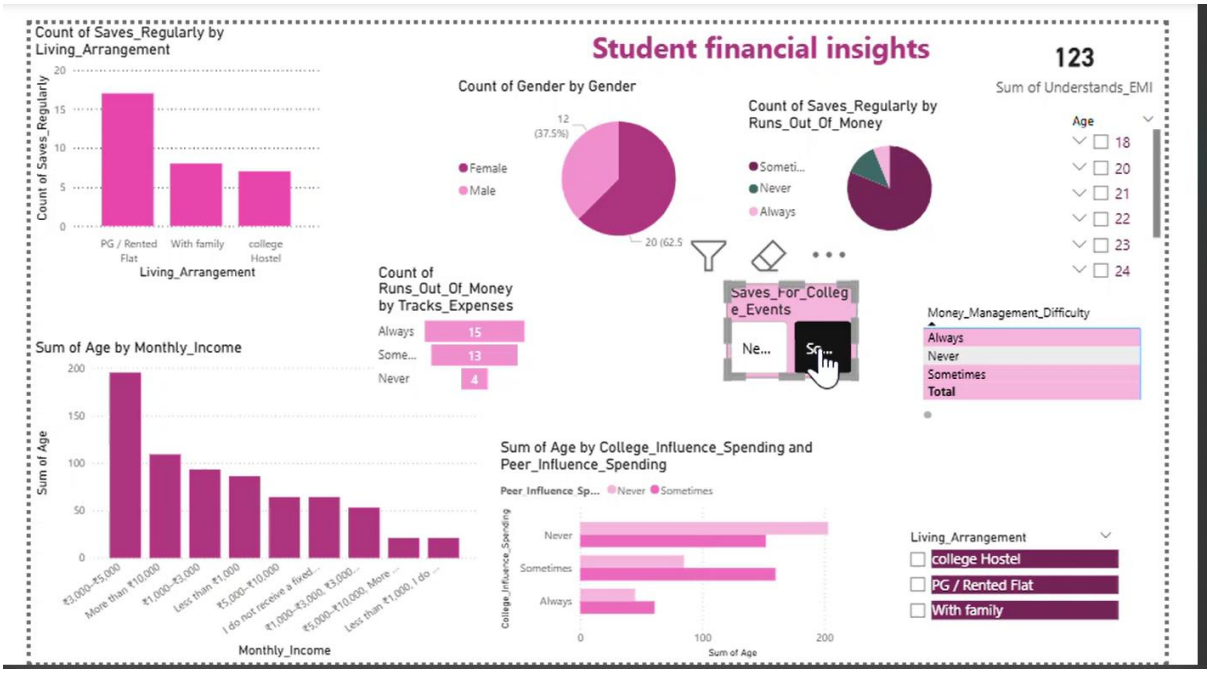
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#The Power BI dashboard clearly indicates that students show mixed levels of financial awareness, with noticeable differences across living arrangements, gender, and income levels. A majority of students save regularly, especially those staying with family and those receiving higher pocket money. However, a significant number of students also report that they sometimes or always run out of money, which reveals inconsistent budgeting and a lack of financial discipline.

#The visual showing expense tracking suggests that students who always track their expenses are less likely to run out of money,

highlighting the importance of awareness and planning. Peer and college influence on spending also affects financial behaviour—students influenced “sometimes” tend to spend more and save less.

#Gender distribution shows more female respondents, but saving habits do not differ significantly between genders. Students' understanding of concepts like EMI appears limited, indicating a gap in deeper financial knowledge.

#Overall, the dashboard suggests that while students show some positive saving habits, financial literacy levels need significant improvement, particularly in budgeting, expense tracking, and understanding financial concepts. Introducing structured financial education programs in colleges could significantly improve students' money-management skills and long-term financial stability.



1. Students strongly prefer digital payments, showing high digital financial literacy.

2. Saving habits are moderate, with many students saving only sometimes.

3. Pocket money amount influences savings—higher allowance = better saving.

4. Food and transportation are the biggest expenses for students.

5. Most students stay with family, which impacts their expenses and saving potential.

6. Students staying in PG/hostel tend to have higher expenses and find it harder to save.

Reliability and Scale Properties

Cronbach's alpha (internal consistency).

The nine-item questionnaire (items listed below) achieved Cronbach's alpha = 0.715 (α

standardized = 0.707; N items = 9; N valid = 95). This level indicates acceptable

internal consistency for group-level research.

Items (full wording)

1. I know the difference between debit cards and credit cards.

2. I understand the importance of having a savings account.

3. I know about digital payment systems (UPI, online wallets, etc.).

4. I have basic knowledge about different saving or investment options (FD, SIP,

mutual funds, etc.).

5. I understand how loan EMIs are calculated and how interest affects the total

amount I repay.

6. I try to save a part of whatever money I receive (pocket money, stipend, etc.).

7. I prefer digital payments over cash.

8. How confident are you in managing money during college days without asking

parents for extra funds?

9. How often do you compare prices (online or in shops) before buying something?

Item–total statistics (selected):

• Highest corrected item–total correlation: I understand the importance of having

a savings account. — r = .705.

• Lowest corrected item–total correlation: I prefer digital payments over cash. — r

= .155.

• Cronbach's alpha if specific items deleted ranged from 0.621 to 0.738; deleting

EMI knowledge would increase alpha to 0.738, showing it contributes less to

internal consistency than others.

Intraclass Correlation Coefficients (ICC):

• Single measures ICC = 0.218 (95% CI: .154–.299), F = 3.507, p < .001.

• Average measures ICC = 0.715 (95% CI: .620–.794), p < .001.

These results agree with Cronbach's alpha and indicate better reliability when

using aggregated (mean) scores.

Hotelling's T^2 and ANOVA:

• Hotelling's T^2 = 522.787, F(8,87) = 60.482, p < .001 — indicating item means

differ significantly across items.

• Within-items ANOVA between-items F ≈ 118.877, p < .001 (per Tukey output) —

further supporting item mean differences.

Correlations

Pearson correlations (selected, N = 95)

Key statistically significant correlations (two-tailed):

• I know the difference between debit and credit cards ↔ I understand the

importance of having a savings account: r = .577, p < .001.

• I understand the importance of having a savings account ↔ I know about digital

payment systems: r = .598, p < .001.

• I have basic knowledge about saving/investment options ↔ I understand how

loan EMIs are calculated: r = .321, p = .001.

• I prefer digital payments over cash ↔ How confident are you in managing

money…: r = .254, p = .013.

Several knowledge items are inter-correlated at moderate to high levels (r ≈ .44–.60; all

p < .001), indicating coherent knowledge structure. Behavioural items show weaker

correlations with knowledge items.

Spearman correlations (rank-order; N = 95)

Spearman's rho confirms Pearson results with rank-based coefficients:

• Knowledge cluster correlations range ρ ≈ .353 to .550, p < .001.

• I try to save part of whatever money I receive correlates with EMI understanding

(ρ = .380, p < .001) and with I prefer digital payments over cash (ρ = .268, p =

.009).

• Price-comparison items correlate modestly with digital preference and

confidence (ρ ≈ .189–.263), but weakly with knowledge items.

Interpretation (summary): Strong associations within knowledge items; weaker and

more varied associations among behaviour/confidence items and price-checking.

Exploratory Factor Analysis (PCA with Varimax rotation)

Factorability checks:

• Kaiser–Meyer–Olkin (KMO) = 0.726 (adequate).

• Bartlett's test of sphericity: $\chi^2(36) = 189.043$, p < .001 (matrix suitable for factor

analysis).

Extraction: Principal components, Varimax rotation. Three components extracted

(eigenvalues > 1), cumulative variance explained = 61.50% (Component 1 = 32.877%,

Component 2 = 16.018%, Component 3 = 12.602% prior to rotation). After rotation, the

rotated sums of squared loadings were: Factor1 ≈ 2.622 (29.13%), Factor2 ≈ 1.676

(18.63%), Factor3 ≈ 1.236 (13.74%).

Rotated component matrix (key loadings) — using your question labels mapped from

Q numbers in your Power BI table. The values below are rotated loadings as reported:

(These correspond to your heatmap table: Total column shows factor-summed absolute

loadings; totals: Factor1 = 3.69, Factor2 = 2.76, Factor3 = 1.70, Total = 8.15).

Communalities (extraction): ranged from .351 (Q9) to .838 (Q5 — EMI understanding).

Interpretation: The rotated solution suggests three interpretable groupings: a strong

knowledge factor (Q1, Q2, Q3, Q4), a behaviour/confidence factor (Q7, Q8, Q9, and

Q6 to some extent), and a loan/EMI factor (Q5 and cross-loading Q6). Factor loadings

are substantial (> .50 on primary loadings for most items), and cross-loading items (Q6,

Q9) are noted where applicable.

question	Factor1	Factor2	Factor3	Total
Q2	0.82	0.29	0.08	**1.18**
Q4	0.71	0.18	0.25	**1.15**
Q6	-0.05	0.56	0.59	**1.10**
Q5	0.24	-0.16	0.87	**0.95**
Q1	0.80	-0.01	0.03	**0.83**
Q3	0.81	0.00	-0.01	**0.80**
Q8	0.23	0.73	-0.22	**0.74**
Q9	0.17	0.57	-0.02	**0.72**
Q7	-0.05	0.61	0.12	**0.69**
Total	**3.69**	**2.76**	**1.70**	**8.15**

Regression Analyses

Regression predicting Price comparison behaviour (dependent = How often do you

compare prices… — VAR00008)

Model fit: $F(8,86) = 59$, $p < .001$; $R = .533$, $R^2 = .284$, Adjusted $R^2 = .217$. Model explains

28.4% of variance in price-comparison.

Coefficients (unstandardized B, SE, standardized β, p):

Predictor (item)	B	SE	β	p
(Constant)	0.568	0.472	—	.232
I know the difference between debit/credit cards	-0.004	0.075	-0.007	.955
I understand the importance of having a savings account	**0.252**	0.090	**0.375**	**.007**
I know about digital payment systems	-0.097	0.081	-0.144	.236
I have basic knowledge about saving/investment options	0.095	0.079	0.142	.231
I understand how loan EMIs are calculated	**-0.149**	0.052	**-0.286**	**.006**
I prefer digital payments over cash	**0.228**	0.113	**0.198**	**.047**
I try to save a part of whatever money I receive	0.197	0.108	0.171	.071
How often do you compare prices (VAR00009) included as predictor?	0.020	0.094	0.021	.829

Collinearity diagnostics: Tolerance values range ~0.459–0.944; VIF values ranged from

~1.06 to 2.18, indicating no serious multicollinearity.

Interpretation: Greater agreement with the importance of having a savings account and

a preference for digital payments are associated with higher reported price-comparison

frequency. Surprisingly, EMI understanding shows a significant negative relationship

with price-comparison behaviour in this model (B = -0.149, p = .006). This suggests the

role of EMI knowledge may be complex — possibly reflecting different financial priorities

or familiarity with credit rather than price sensitivity.

Regression predicting VAR00012

The regression predicting VAR00012 returned R = 1.000 and R^2 = 1.000 with zero

residual variance and identical coefficients (B ≈ .111). This indicates that VAR00012 is

a deterministic linear combination (likely computed as a sum/average of the

predictors) or there is a data/configuration artifact. This model is not interpretable as

inferential regression and is excluded from substantive interpretation.

Independent-Samples t-Test and Effect Sizes

An independent-samples t-test compared group means of VAR00012 between two

groups (Group 1, n = 58; Group 2, n = 35).

Descriptive:

• Group 1 mean = 3.3697, SD = .4936

• Group 2 mean = 3.3175, SD = .5768

Levene's test for equality of variances: F = 1.922, p = .169 (variances assumed equal).

t-test result: t(91) ≈ 0.464, two-tailed p = .644 (not significant).

Effect sizes (SPSS table):

• Cohen's d = 0.526 (point estimate) ; 95% CI: −0.321 to 0.519.

• Hedges' g = 0.531 ; 95% CI: −0.318 to 0.515.

• Glass's Δ = 0.577 ; 95% CI: −0.330 to 0.510.

Interpretation: Although point estimates indicate a moderate effect size (d ≈ 0.53), all

95% confidence intervals include zero; therefore the difference is not statistically

reliable. Practically, group differences are small and not consistent.

Summary of Detailed Results

• The nine-item measure is reliable (α = .715; ICC average = .715).

• Correlation analyses show coherent relationships among knowledge items and

weaker links between knowledge and behaviour.

• EFA (PCA) extracted three components explaining 61.50% of variance; rotated

loadings show a knowledge cluster, a behaviour/confidence cluster, and an EMI

specific component.

• Regression predicting price-comparison behaviour is significant (R^2 = .284), with

savings-account importance (positive), EMI understanding (negative), and digital

preference (positive) as significant predictors.

• Between-group t-test shows no significant mean differences, although effect

size estimates are moderate but not statistically reliable.

• Chi-square tests mostly non-significant except VAR00017 × VAR00018 which is

significant but affected by small cell counts.

Overview of Main Findings

This study examined students' self-reported financial knowledge, behaviour, and

decision-making practices. The questionnaire demonstrated acceptable internal

consistency (α = .715). Analyses provide a nuanced picture: students self-report strong

knowledge of financial concepts (debit/credit, digital payments, savings), yet reported

financial behaviours and confidence (saving, price comparison, self-management) lag

behind.

Interpretation of Key Results

Knowledge vs Behaviour gap.

Consistent correlations among knowledge items (r up to .598) indicate students who

understand one financial topic tend to understand others. However, weaker

associations between these knowledge items and practical behaviours (e.g., saving

frequency, price comparison) demonstrate a knowledge–action gap: knowledge alone

is insufficient to predict responsible financial practices.

Factor structure.

The PCA yielded three components, corroborating the multi-dimensionality of financial

literacy in this sample: (1) conceptual knowledge about banking and payment systems;

(2) behaviour and decision-making tendencies (digital preference, price comparison,

confidence); and (3) EMI/loan-specific understanding. The EMI item loading on a

separate component indicates that understanding of credit/interest mechanics forms a

distinct domain that may require separate instructional emphasis.

Predictors of price comparison.

Multiple regression found that perceived importance of savings and preference for

digital payments positively predicted frequency of price checking, while EMI knowledge

predicted less frequent price comparison. One interpretation: students who are

savings-oriented and tech-savvy are more price-conscious; students with EMI

knowledge may focus on credit structures rather than small-item price competition.

Alternatively, the negative EMI coefficient might reflect a subgroup with higher technical

knowledge but lower day-to-day price-sensitivity.

Group differences.

No statistically significant group differences were found for VAR00012 across the two

groups examined. Effect-size estimates suggested moderate differences but were not

statistically reliable (95% CIs included zero), indicating that any observed group mean

differences are inconclusive.

Categorical associations.

Most chi-square tests were non-significant, indicating distributions across categories

did not differ meaningfully. The significant VAR00017 × VAR00018 result is unreliable

due to small expected cell counts and should be investigated with larger samples or

collapsed categories.

Practical implications

• Financial-education programs should not only convey knowledge but also

include behavioural interventions (habit formation, budgeting practice, nudges)

to bridge the knowledge–action gap.

• EMI/loan mechanics appear as a specialized domain — curriculum modules on

credit literacy may be helpful.

• Digital-payment familiarity is high; leveraging digital tools (apps for budget

tracking/price comparison) might improve money-management behaviours.

Conclusions

The study's instrument is a valid and reliable measure of several dimensions of

students' financial literacy. Students report high knowledge but lower behaviouralenactment and confidence. Financial literacy in this sample is multi-dimensional

(knowledge, behaviour, EMI knowledge), and these components relate in predictable

but non-identical ways.

Practical Recommendations

1. Design applied workshops that combine knowledge with hands-on budgeting

exercises and price-comparison tasks.

2. Develop EMI/credit literacy modules emphasizing calculation practice and

consequences of interest rates.

3. Use digital interventions (mobile budgeting tools, price-comparison browser

extensions/apps) to translate knowledge into behaviour.

4. Measure outcomes with pre-post designs to evaluate whether knowledge gains

translate to behaviour change.

Limitations

• Sample size (~95) limits power for some subgroup analyses and produces small

expected counts in some chi-square tables.

• Self-report measures may be subject to social desirability or recall bias.

• One regression (VAR00012) was deterministic, indicating a computed dependent

variable — such constructed variables should be used cautiously in inferential

models.

• Cross-sectional design cannot support causal claims.

Future Research

• Longitudinal studies to assess whether knowledge predicts behaviour change

over time.

• Experimental interventions (randomized controlled trials) to test the

effectiveness of behavioural nudges, apps, or financial coaching.

• Larger, more diverse samples to support reliable chi-square analyses and

subgroup comparisons.